Forex Market Part 1

Financial Institutions and Markets

Topics

- Importance of forex markets
- Exchange rates
- How forex rates are calculated
- Important forex currencies and locations
- Purchase power parity

Activity

What would happen without forex markets?

Life without forex markets

- Trade between nations would require universal money
  - Like the gold standard
- There would be no carry trade
  - Borrowing USD, trade for NZD to invest in NZD bonds
- There would be fewer opportunities for speculation
  - Like gambling, so question if this is good
- The forex market is where exchange rates are set.

Types of forex markets

- **Spot**: Immediate pricing, immediate delivery
- **Futures**: Standardized contracts
  - Limited dates, amounts, etc.
- **Forwards**: Custom contracts (as needed)
- We will focus on the spot market.

US Dollar	1	
Euro	0.86435	-1%
British Pound	0.74497	-1.3%
Japanese Yen	158.48	-0.47%
Canadian Dollar	1.3718	-0.21%
Vietnamese Dong	26,299	-0.084%
Gold Ounce	0.00021305	+0.28%

Forex rates change over time

bid rate	*offer rate*
↓	↓
US$/£ 1.5986	1.5990
↑	↑
You can buy dollars from a bank or broker at this rate.	You can sell dollars to a bank or broker at this rate.

The difference (the spread) between bid and offer is 4 pips

if you wished to purchase US$1 million the cost would be:

$$\frac{\$1{,}000{,}000}{1.5986} = £625{,}547$$

if you wished to sell US$1 million you would receive:

$$\frac{\$1{,}000{,}000}{1.5990} = £625{,}391$$

Forex rates of £

Currency	£1 7-Jul-16	£1 52 wk-high	£1 52 wk-low
Australian Dollar	1.7254	2.2037	1.7189
Canadian Dollar	1.6728	2.0847	1.6728
Chinese Yuan	8.6373	10.0769	8.6201
Czech Koruna	31.5907	38.9637	31.4705
Danish Krone	8.6916	10.7461	8.6607
Euro	1.168	1.4402	1.1639
Hong Kong Dollar	10.0338	12.204	9.9976
Indian Rupee	87.2112	105.136	86.9703
Japanese Yen	130.3287	195.0906	129.9066
Malaysian ringgit	5.2051	6.75	5.1862
New Zealand Dollar	1.7921	2.4596	1.7921
Norwegian Krone	10.9538	13.2607	10.9385
Polish Zloty	5.1682	6.0886	5.1682
Russian Ruble	82.7374	118.8541	82.7374
Saudi Riyal	4.8516	5.9027	4.8339
Singapore Dollar	1.7444	2.2162	1.7421
South African Rand	19.0402	24.3647	19.0402
South Korean Won	1494.6094	1874.4115	1494.6094
Swedish Krona	11.091	13.6302	11.0092
Swiss Franc	1.264	1.5514	1.2596

The highs and lows are more extreme than usual because of the Brexit vote.

Source: BoE, 2016

Activity

Identify (or guess) the top 10 currencies in foreign exchange markets?

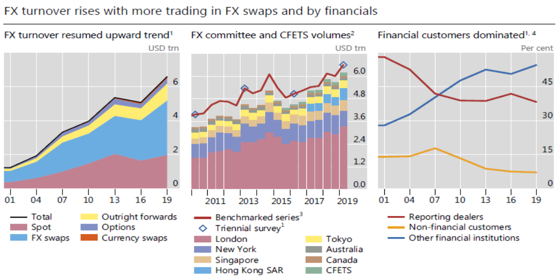
Hint: VND is not in the top 10.

1. USD
2. EUR
3. JPY
4. GBP
5. CNY
6. AUD
7. CAD
8. CHF
9. HKD
10. SGD

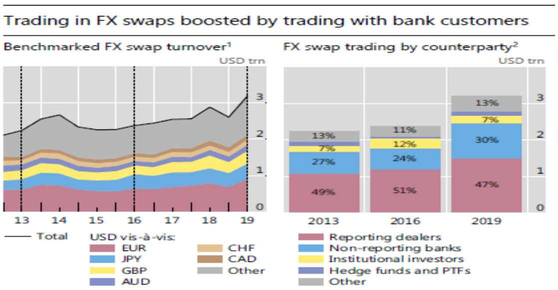
Activity

Identify which geographic locations trade the most forex?

Hint: Hanoi is not in the top 5.



Source: Schrimpf and Sushko (2019)



Source: Schrimpf and Sushko (2019)

Participants in forex markets

- Commercial companies with export or import interests
- International banks and commercial banks carrying our proprietary trading
- Dealers trading and acting as market makers in currency (usually the big banks)
- Brokers buying and selling currencies on behalf of a client, e.g. a client bank or commercial firm
- Governments needing foreign currency for overseas trade or to pay for activities abroad
- Speculators or arbitrageurs taking advantage of perceived FX pricing anomalies
- Tourists or investors (e.g. in property or shares) needing to pay in foreign currency
- Fund managers investing abroad (pensions, insurance companies, etc.)
- Central banks (smoothing out fluctuations or managing the rate to a desired level).

Activity

Identify which factors could influence the supply or demand for the VN dong.

Purchasing power parity (PPP)

- Basket A (US) = Basket B (UK)
- If Basket A = $357 and Basket B = £229
  - Exchange rate should be $1.56 = £1
- If different, arbitrage.

PPP

- US inflation = 3%, Basket A = $15,000
- UK inflation = 15%, Basket B = £10,000
- PPP equilibrium: £1 = $1.50
- What about 1 year later?

Critical of PPP

- Goods must be freely traded, no transaction cost
- Works in long run (may take years)
  - Customers slow to recognize incentive
  - Gov't may manage forex rates
  - Balance of payments disequilibrium
  - Capital transactions
  - Speculation
  - Inflation is only one factor
  - 90% of forex unrelated to trade (it just speculation because it moves so fast

Arbitrage

- The purchase of commodities in markets where they are cheap for sale in markets where they are expensive
- If the £-€ exchange rate is £1 = €2 in London and £1 = €1.90 in Singapore, arbitrageurs would profit by buying Euros for pounds in London and simultaneously selling Euros for pounds in Singapore
- e.g. an arbitrageur could buy €2,000 for £1,000 in London and immediately sell €2,000 for £1,052.63 in Singapore, earning a 5.26% profit
- This activity would cause the demand for Euros in London to rise – making Euros more expensive in London – and the supply of Euros in Singapore to rise – making Euros cheaper in Singapore – until the exchange rates became equal in both centers.

Arbitrage (2)

- Since currencies are traded in markets like any other commodity, their prices – exchange rates – will rise or fall in response to changes in supply and demand conditions
- Where exchange rates are allowed to vary in this way they are said to be floating
- Often, the currency departments of central banks intervene in the forex markets to fix the exchange rate of their national currency at some preferred value
- If, for example, the Bank of England wanted Sterling fixed at some particular exchange rate against the Euro, it would sell pounds if Sterling rose in value against the Euro and buy pounds when sterling's value against the Euro was falling.

Review

Key Points:

- Exchange rates are the prices of currencies
- Rates change with supply and demand
- PPP helps predict the future